

CONFIDENTIAL // SOVEREIGN ARCHITECTURE SPECIFICATION

SOVEREIGN INTELLIGENCE HUB

Autonomous Systems Master Plan & AI Deployment Guide for Apple Silicon Mac Mini

HOST PLATFORM	CORE RUNTIME	TARGET OPERATOR	DOCUMENT VERSION
Apple Silicon Mac Mini (M-Series)	OpenClaw Gateway + Local MLX	Founder, Survivalist Squads / NYC	v3.4 Production Master (2026)



INGESTION DIRECTIVE FOR YOUR LOCAL OPENCLAW AGENT

If you are feeding this document into your OpenClaw agent, copy and paste this command block to generate your interactive briefing:

```
"OpenClaw: Ingest this Master Plan. You are my Sovereign Intelligence Architect.
1. Provide a 3-minute executive briefing of the 6 core modules.
2. Generate an interactive selection menu so I can choose which modules to activate (e.g. 1-6).
3. For any module I select, produce the step-by-step install script, configuration plist, and verification check."
```

1. Executive Summary & Core Architectural Thesis

Most AI agents operate reactively as cloud chatbots with shell access. The **Sovereign Intelligence Hub** inverts this model: it turns an always-on, low-power (~10W) Apple Silicon Mac Mini into a **proactive, unblinking local intelligence station**.

By leveraging Apple Silicon's **Zero-Copy Unified Memory Architecture (UMA)** and on-device **Neural Engine (ANE)**, the node runs continuous digital signal processing, sub-second speech-to-text, real-time computer vision, relational graph traversals, and off-grid radio mesh operations without sending private data to cloud servers.

MODULE 1: The Autonomous Relational Graph & Social Arbitrage Engine

RELATIONAL CAPITAL // ZERO LEAK

Strategic Objective: Transform an extensive network of 1,000+ contacts (C-suite, founders, old friends, dating circles) into an active, compounding intelligence asset.

- Temporal Decay Formula** ($SS_i(t) = \sum w_k e^{-\lambda \Delta t}$): Tracks relationship decay dynamically across tiers (VIP $\lambda=0.007$, Social $\lambda=0.046$). Auto-stages witty, contextual check-in drafts before connections go cold.
- Structural Hole Arbitrage:** Background vector traversals match unlinked nodes whose active *Needs* semantically match another's *Assets* ($\text{Cosine Sim} \geq 0.82$). Stages power introductions for the Operator.
- AirPod Pre-Meeting Audio Dossier ("The Uber Briefing"):** Drop a name or business card photo while en route; OpenClaw synthesizes a 90-second CIA-style audio debrief into your AirPods with psychological cues and conversational landmines.
- Psycholinguistic Chameleon Engine:** Mathematically extracts a 5-point stylistic entropy vector from incoming messages (formality, humor, lexical diversity) to dynamically mirror the recipient's tone in all ghostwritten drafts.

MODULE 2: The Den Orchestrator & AI Parlor Tricks

ATMOSPHERE // INTERACTIVE MENTALISM

Strategic Objective: Turn the host's intimate after-party den into a responsive, magic-infused social stage using ambient camera vision and Alexa smart lighting.

- **The AI "Mentalist" Cold-Read:** Room camera processes physical items (books, cards, vintage watches) via local sub-second vision models, whispering exact trivia and historical details directly to the Operator's Apple Watch.
- **Subliminal Story-Coded Lighting:** Local Whisper monitors storytelling for natural trigger phrases (*"and then the room went pitch black"*, *"welcome to the inner sanctum"*), shifting Alexa lighting scenes seamlessly without touching an app.
- **Vision-Based Remote Lie Detector:** Tracks microscopic facial blood flow pulses (Remote Photoplethysmography - rPPG) and micro-saccades during party games (*Two Truths and a Lie*), sending subtle haptic alerts on pulse spikes.
- **The Cocktail Oracle:** Tangible object detection identifies bottles/records placed on the coffee table; the speakers announce bespoke drink recipes in an omniscient British butler voice while lighting shifts to match.

MODULE 3: Multi-Spectrum SIGINT, HAM Radio & Dual-Path Alerting

SIGNALS INTELLIGENCE // REAL-TIME
DISPATCH

Strategic Objective: Passively ingest local radio frequency spectrum, emergency dispatch, and amateur nets, translating incidents with sub-second latency.

- **Hardware Ingestion:** Connects a dual-band HAM radio (via Digirig USB) or RTL-SDR dongle to capture local repeaters, APRS packets (144.390 MHz), and unencrypted P25 police/fire talkgroups.
- **Neural Voice Squelch & Whisper:** Silero-VAD triggers CoreML Whisper on the Neural Engine in $<250\text{ms}$, translating 10-codes (`10-75` working fire, `10-13` officer down) and geocoding cross-streets.
- **Dual-Path Alert Dispatch:**
 - *Path A (Cellular/Internet):* Rich Telegram/WhatsApp alert with audio clip, transcript, and Google Maps pin.
 - *Path B (Off-Grid LoRa):* Compact 240-byte binary packet beamed over 915 MHz to a pocket Meshtastic radio across the city with zero internet.
- **Acoustic Event Detection:** Window boundary microphones running CoreML YAMNet classify sirens, breaking glass, and gunshots in real time.

MODULE 4: Sovereign Off-Grid $\mathbb{C}^4\text{I}$ Mesh (*Survivalist Squads*)

CRISIS RESILIENCE // RETICULUM MESH

Strategic Objective: Build an un-killable, decentralized command and logistics hub for Survivalist Squads that operates during complete infrastructure collapse.

- **915 MHz Reticulum Network Protocol:** Zero-config cryptographic routing utilizing Ed25519 digital signatures and X25519 ECDH encryption over pocket LoRa radios.
- **Air-Gapped Neural Knowledge Base:** Integrates local 4-bit MLX LLMs with offline Kiwix trauma surgery manuals, FEMA disaster checklists, and OpenStreetMap vector tiles.
- **Cryptographic CRDT Logistics Ledger:** Tracks distributed medical kits, water, fuel, and comms assets across squad members using Conflict-Free Replicated Data Types (LWW-Element-Sets) over asynchronous radio hops.

MODULE 5: Spycraft, OSINT & Counter-Surveillance Tradecraft

ALL-SOURCE RECON // DEFENSE

Strategic Objective: Equip the Operator with nation-state grade target reconnaissance, HUMINT elicitation tools, and physical counter-surveillance.

- **Autonomous Target "Red Files":** Automated scrapers query court dockets (RECAP/PACER), SEC EDGAR filings, property mortgages (ACRIS), and timestamp Pattern-of-Life models to build leverage profiles.
- **HUMINT Elicitation Playbooks:** Pre-dinner scripts (Provocative Falsehoods, Mutual Grievances, Feigned Naivety) calibrated to make targets volunteer non-public deal terms.
- **Canary Trap Watermarking:** Embeds invisible zero-width unicode characters and synonym permutations in distributed decks to mathematically attribute leaks.
- **Chaperone Tail & Stingray Defense:** Correlates BLE/Wi-Fi probe requests across different neighborhoods to detect persistent physical surveillance tails and monitors cellular BCCH channels for rogue IMSI catchers.

MODULE 6: Manhattan Urban Alpha & Scarcity Sniping

LOCAL ARBITRAGE // VIP ACCESS

Strategic Objective: Extract hidden economic value, early real estate intelligence, and scarce hospitality access from high-density Manhattan.

- **NYC Municipal Alpha Engine:** Mines daily State Liquor Authority (SLA) filings and Department of Buildings (DOB) alteration permits (> \$250k) to predict exclusive new speakeasies and clubs 3–9 months before opening.
- **Autonomous Reservation Sniper:** Headless browser agents monitor Resy/SevenRooms cancellation drops for impossible 8:00 PM tables (4 Charles, Torrisi, Polo Bar) on demand.
- **Secondary Luxury Asset Scanner:** Scrapes Chrono24 and fine art auction catalogs, flagging underpriced references (Daytona/Patek) against 30-day moving averages.

2. Hardware Bill of Materials (BOM) & Architecture Map

Component / Subsystem	Hardware Specification	Estimated Cost	Key Software / Libraries
Core Sovereign Node	Apple Silicon Mac Mini (M2/M3/M4, 16GB+ Unified Memory)	Existing / \$599	macOS launchd, OpenClaw, MLX, CoreML
Off-Grid Mesh Transceiver	Heltec V3 / LilyGO T-Beam (915 MHz LoRa ESP32)	\$25 - \$35	Meshtastic Firmware, Reticulum (RNS), PySerial
RF Spectrum Scanner	RTL-SDR Blog V4 Dongle + Multi-Band Dipole Antenna	\$35	rtl_433, dump1090-fa, op25, acarsdec
HAM Radio Digital Interface	Digirig Mobile (USB Audio/PTT) + Baofeng / Yaesu Cable	\$40 - \$50	direwolf, soundmodem, Silero-VAD
Acoustic & TSCM Sensor	Calibrated USB Boundary Microphone (Window / Terrace)	\$30 - \$60	CoreML YAMNet, PyAudio, CLAP Audio Classifier
Den Vision & Atmosphere	RTSP / IP Room Camera + Alexa Smart Lights (Hue/Lutron)	Existing / Variable	Moondream VLM, OpenCV rPPG, Homebridge / Alexa API
Uninterruptible Power	Mini-UPS Battery Backup (12V/120V Pass-Through)	\$45 - \$80	Automated Power-Fail Keep-Alive Scripts

3. Phased Implementation Roadmap

Phase	Focus Area	Key Milestones	Validation Gate
Phase 1: Foundation (Day 1-3)	Sovereign Daemon Core & PRM	Configure launchd daemons, local SQLite vector graph, and WhatsApp/Telegram gateway.	Send 30s voice memo; verify entity extraction and auto-drafted follow-up in Drafts.
Phase 2: Signals & Alerting (Day 4-7)	SIGINT, HAM Radio & LoRa	Plug in RTL-SDR & LoRa node. Setup Silero-VAD + CoreML Whisper pipeline.	Simulate P25 fire dispatch; verify dual alert receipt on Telegram and pocket LoRa radio.
Phase 3: The Den & Hospitality (Day 8-10)	Vision Mentalism & Smart Lighting	Integrate room RTSP camera stream, rPPG pulse tracker, and story-coded lighting cues.	Speak test keyword; confirm instant silent lighting shift and Apple Watch whisper.
Phase 4: Spycraft & Squads (Day 11-14)	OSINT Red Files & CRDT Mesh	Deploy RECAP court scrapers, canary watermarkers, and Reticulum offline logistics.	Run automated target recon; verify 1-page dossier generation within 60 seconds.

4. Interactive Agent Selection Menu (For Operator)

When ready to proceed, simply tell your OpenClaw agent which modules or combinations you want to bring online:

- [OPTION 1] **The Executive Network Bundle:** Module 1 (Relational Graph) + Module 6 (Manhattan Scarcity Sniper)
- [OPTION 2] **The Tactical Squad Node:** Module 3 (HAM / Emergency SIGINT) + Module 4 (Off-Grid LoRa Mesh)
- [OPTION 3] **The Den Mastermind:** Module 2 (AI Parlor Tricks & Lighting) + Module 5 (OSINT Red Files & Elicitation)
- [OPTION 4] **Full Sovereign Deployment:** Activate all 6 modules simultaneously.

Security Note: All operational configurations, vector embeddings, and RF captures reside strictly on local NVMe storage within the physical perimeter of the Mac Mini. Zero unencrypted telemetry is transmitted to external cloud infrastructure.